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# Federated Learning for Diabetes Prediction: Investigating Client-Level Statistical Heterogeneity

**Abstract**—Diabetes prediction using machine learning can support early identification of individuals at risk, however, existing centralized methods rely on patient data being collected and aggregated into a central repository. This raises data sharing and data governance issues for healthcare organizations that must cooperate for this purpose. Another approach is federated learning in which a joint model is trained cooperatively by healthcare organizations and raw data does not leave the respective organization. Although the performance of federated learning depends on the degree of class imbalance, differences in data distributions on the side of the clients, as well as the number of clients, its exact effect is yet unknown. This research explores the above factors via a comparative study of centralized and federated learning for predicting diabetes. Logistic regression is the base prediction model used, while class imbalance is solved by SMOTENC algorithm. In federated learning, local SMOTENC technique is performed separately for each client, local training is performed using FedProx, and server aggregation is performed using sample size weighted average aggregation technique. Evaluation is performed for both IID and non-IID client data distributions, where non-IID client data distributions are performed using Dirichlet distribution ( $\alpha = 0.30$ ). The study also examines the impact of the change of the number of clients from 3 to 5 and 10. Evaluation of the resulting models is done on the same unseen testing set, using the accuracy, precision, recall, F1 score, ROC-AUC, and PR-AUC metrics. The outcomes allow comparing the performance of centralized and federated approaches to diabetes prediction.

**Keywords**—federated learning, diabetes prediction, logistic regression, class imbalance, SMOTENC, FedProx, client heterogeneity

## I. INTRODUCTION

Diabetes mellitus is a chronic disease that affected an estimated 828 million adults worldwide in 2022 [11] and can lead to serious complications such as cardiovascular disease, renal failure, and diabetic retinopathy. Early and accurate prediction of diabetes is therefore of considerable importance, since it can support timely intervention, better disease management, and healthcare planning [1]. With the growing availability and use of healthcare data, machine learning techniques are increasingly applied to identify patterns in clinical information and support disease prediction and decision-making. The growth in medical data also brings challenges in computation, security, governance, and cross-institutional sharing, particularly when information comes from multiple healthcare institutions [4].

The use of centralized machine learning requires patient records from different institutions to be collected in a common repository before training. Although this approach allows models to learn from a larger and more diverse dataset, healthcare data is highly sensitive and subject to restrictions related to privacy, governance, so centralizing it creates security risks. Healthcare institutions may therefore be unable or unwilling to share raw patient records [2], creating a significant challenge for machine learning which benefits from diverse data yet patient information is heavily

governed by institutional regulations meaning data cannot be seamlessly transferred between organizations.

Federated learning provides a distributed solution to this issue by enabling several healthcare organizations to jointly learn a model without sharing any data at all. Instead of gathering data into one place, the participants will be training models locally and will only share the output, i.e., the updates to the model [3]. Such a method has attracted much interest in medicine since it enables the organization to learn jointly while keeping their patients' information secure. Thus, federated learning has been used for some clinical applications, such as diabetes prediction and electronic health record analysis [6].

Related diabetes prediction studies have used techniques such as the Synthetic Minority Over-sampling Technique (SMOTE) to address class imbalance and have compared different machine learning and federated learning approaches [1]. Client-level oversampling combined with FedProx has also been applied to imbalanced cardiovascular risk data, where FedProx outperformed standard FedAvg [13]. In practice, client data is often non-IID: class proportions and feature distributions differ across institutions, which can affect convergence and accuracy in federated training [12]. Federated learning research in healthcare has also examined distributed prediction with different numbers of clients, showing that data distribution and system configuration affect model performance [3]. Although further investigation is required in understanding how centralized learning and federated learning compare when the same prediction methodology is maintained, few studies have jointly considered factors such as class imbalance, IID and non-IID data distributions, and changes in the number of participating clients.

This study therefore investigates how client-level statistical heterogeneity affects federated learning performance for diabetes prediction and compares the resulting federated model with an equivalent centralized model. More precisely, it examines how heterogeneity in client data affects predictive performance relative to centralized learning, and how the gap between the two differs between IID and non-IID client distributions. It also evaluates 3, 5, and 10 clients under both settings to see how federation size and data distribution affect performance. Logistic regression is the prediction model, SMOTENC (Synthetic Minority Over-sampling Technique for Nominal and Continuous features) handles class imbalance, and FedProx is used for federated training. A single classification threshold is selected once, using three-fold out-of-fold predictions from the centralized training data to maximize F1-score, and is then fixed across all centralized and federated experiments. Performance is evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC.

## II. LITERATURE SURVEY

Dash et al. [1] study privacy-preserving methods of predicting diabetes and heart diseases based on federated learning due to the necessity of early diabetes diagnosis and difficulties with data exchange in the health-care sector in the context of centralized machine learning. FL with Weighted Conglomeration Optimization (FLWCO) method by the authors is aimed at selecting local classifiers according to their performance level and using confidence score thresholds, which improves communication efficiency and convergence rate.

Grant et al. [2] compare federated learning with centralized learning for diabetic classification with respect to the PIMA Indians Diabetes Dataset, where they apply logistic regression to determine if federated learning is able to equal centralized learning without requiring central collection of the raw data. Federated learning achieves ROC-AUC score of 0.77, which is close to the baseline of 0.78 achieved with centralized learning, making their paper an important reference point in comparing these methods with structured diabetes data. However, the authors do not consider the problem of non-IID data distribution.

Haripriya et al. [3] present a Federated Learning approach with adaptive aggregation for privacy-enabled image classification in medical images. With the use of chest X-rays, brain MRIs, and diabetic retinopathy datasets, the dynamic aggregation method of Haripriya et al., which dynamically changes between FedAvg and FedSGD according to the weight variation of the clients, achieves up to 98.6% accuracy in classifying brain tumors with fine-tuning on EfficientNetV2. This work mainly evaluates its approach through a simulated system consisting of ten clients. Initial tests with up to 100 clients reveal increased communication cost and reduced accuracy, raising the issue of scalability to hundreds of clients.

Demir et al. [4] study privacy-preserving diabetes and heart disease prediction with federated learning, motivated by the privacy, security, and regulatory limits on centralizing medical records. Their Data Unaware Classification Based on Association (duCBA) algorithm aggregates logical relational rules instead of high-dimensional model weights, which improves privacy and computational efficiency while keeping decisions interpretable. Statistical heterogeneity across clients is not addressed.

Atmoko et al. [5] propose FedXChain, an explainable federated learning framework that addresses heterogeneous and non-IID healthcare data through SHAP-based aggregation, adaptive trust-weighted aggregation, and blockchain auditing. On the Wisconsin Breast Cancer dataset, their logistic regression model reaches 96.5% accuracy. The evaluation uses a fixed ten-client setup and compares only against federated baselines (FedAvg and FedProx), so the effect of client count, and any comparison with an equivalent centralized model, is left unexamined.

Across these studies, the federated learning literature for disease prediction has concentrated on developing better aggregation, privacy, or optimization mechanisms, most often under a fixed client configuration. Most include a centralized baseline, but few examine how the gap between centralized and federated performance changes with client data distribution. Among the studies that scale the number of clients, Haripriya et al. and Shahid et al. evaluate non-IID partitions only, and Demir et al. evaluates IID partitions only. No reviewed study compares IID and non-IID partitions across client counts against a centralized baseline while also handling class imbalance. This leaves open how class imbalance, IID versus non-IID client

Bebortta et al. [6] examine privacy preserving prediction of cardiac risks using federated learning on IoT-enabled Electronic Health Records (EHR). The proposed FedEHR approach employs a sparse support vector machine, a cluster primal-dual splitting algorithm, and round-robin heuristic to enhance the predictive ability while cutting down on communication and computation overheads. The paper recognizes data quality, heterogeneity, and scalability as open problems, but it does not analyze the influence of different client-side data distribution on the performance disparity of centralized versus federated learning approaches.

Shahid et al. [7] examine privacy-preserving chronic disease prediction using federated learning, motivated by the need for early diagnosis and the privacy, regulatory, and interoperability challenges of centralizing patient data. Their framework, Health-FedNet, combines calibrated differential privacy, Paillier homomorphic encryption, and adaptive node weighting, improving predictive accuracy, cutting communication overhead by 41.6%, and lowering data-leakage risk.

Rahman et al. [8] target privacy-preserving early stroke risk prediction and continuous patient monitoring through a federated, IoT-enabled machine learning framework. Their approach pairs a Federated Averaging (FedAvg) pipeline with a wearable sensing system, combining multi-stage feature optimization (PSO, mRMR, RFC, and PCA/LDA) with real-time limb-temperature asymmetry and blood-oxygen (SaO<sub>2</sub>) tracking, reaching up to 99.44% accuracy. Because the evaluation relies on a single Kaggle dataset with stratified IID partitions, validation under non-IID client variation is left for future work.

Sorayaie Azar et al. [9] investigate privacy-preserving coronary artery disease prediction using federated learning. Their framework, FMLCA, combines decentralized machine learning classifiers with k-anonymity, implemented via the IBM AI Privacy Toolkit, and SHAP-based explainability to protect patient confidentiality and support clinical trust, reaching 83.21% classification accuracy. The evaluation is limited to a relatively small clinical dataset, so performance across diverse, heterogeneous multi-center settings remains untested.

Amin et al. [10] propose Distribution-Aware Federated Learning (DA-FL) for diabetes prediction on the CDC BRFSS 2021 dataset, addressing non-IID data and class imbalance through class-weighted local training and distribution-aware aggregation. The work centers on the aggregation strategy itself and uses a fixed five-client federation, leaving the effect of varying client numbers, and the resulting performance shifts under IID and non-IID conditions, less explored.

Table 1 summarizes these studies alongside the present work.

distributions, and the number of participating clients jointly affect the gap between centralized and federated performance, which is the focus of the present work.

| Work                 | Dataset                                                                    | FL Method                                                            | IID / Non-IID                                                        | Client Count                     | Imbalance Handling                                                                  | Centralized Baseline                                                 |
|----------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| [1]                  | Kaggle Diabetes (96,146 records)                                           | Weighted Conglomeration Optimization (FLWCO), with FedAvg/FedMAP     | Not specified                                                        | 4 clients                        | SMOTE                                                                               | <b>Yes</b> (Centralized RF, XGBoost, MLP, Gradient Boost, AdaBoost)  |
| [2]                  | PIMA                                                                       | Flower federated Logistic Regression                                 | IID                                                                  | 3 clients                        | None (Evaluated natural class skew)                                                 | <b>Yes</b> (Centralized RF, SVM, LR)                                 |
| [3]                  | Multimodal Medical Images                                                  | Adaptive Aggregation (FedAvg / FedSGD switching) + Transfer Learning | Non-IID (random vs. stratified)                                      | 10 clients (Evaluated up to 100) | None (Pretrained ImageNet weights)                                                  | <b>Yes</b> (Centralized Transfer Learning with DP)                   |
| [4]                  | CDC Diabetes Health Indicators and the Kaggle Heart Attack dataset (Fayez) | duCBA (Data-Unaware Classification Based on Association) vs. FedAvg  | IID                                                                  | 2, 4, 8 clients                  | Random under-sampling on Heart Attack; the diabetes set is already balanced.        | <b>Yes</b> (Centralized CBA)                                         |
| [5]                  | Wisconsin Breast Cancer (WDBC) & Synthetic set                             | FedXChain (Adaptive Trust Scoring + Blockchain + SHAP)               | Both IID & Non-IID (Dirichlet $\alpha = 0.3, 0.5$ )                  | 10 nodes                         | Not specified                                                                       | <b>No</b>                                                            |
| [6]                  | de-identified EHRs from Boston Medical Center                              | FedEHR (Sparse SVM + CPDS + Round-Robin Heuristic)                   | Not specified                                                        | 5 clients                        | Not specified                                                                       | <b>Yes</b> (Centralized SVM)                                         |
| [7]                  | MIMIC-III, diabetes and hypertension                                       | Health-FedNet (Adaptive Node Weighting + DP + Paillier HE)           | Non-IID                                                              | 5 to 100 clients                 | Threshold tuning / None                                                             | <b>Yes</b> (Centralized Deep Learning)                               |
| [8]                  | Kaggle Stroke Risk Prediction Dataset Based on Symptoms (70,000 records)   | FedAvg + Multi-stage Feature Selection (PSO, mRMR, RFC)              | IID (Stratified; Non-IID left as future work)                        | 10 clients (Fixed)               | None (Evaluated via F1 / MCC)                                                       | <b>Yes</b> (Centralized XGBoost, RF, MLP)                            |
| [9]                  | UCI Heart Disease (Cleveland), 303 samples                                 | FMLCA (Federated ML + k-anonymity + SHAP)                            | Not specified                                                        | 5 clients                        | SMOTE                                                                               | <b>No</b>                                                            |
| [10]                 | CDC BRFS 2021 Diabetes (236,378 records)                                   | DA-FL (Distribution-Aware FL + Minority Amplification $\phi_k$ )     | Non-IID (Dirichlet $\alpha \in [0.1, 1.0]$ )                         | 5 clients (Fixed)                | Class-Weighted Cross-Entropy + Minority Amplification                               | <b>Yes</b> (Centralized MLP)                                         |
| <b>Proposed Work</b> | <b>Kaggle Diabetes Prediction Dataset (96,146 records)</b>                 | <b>FedProx (Federated Logistic Regression with Proximal Term)</b>    | <b>Both IID &amp; Non-IID (Dirichlet <math>\alpha = 0.30</math>)</b> | <b>Varied (3, 5, 10 clients)</b> | <b>Local SMOTENC per client + Centralized OOF F1-Maximizing Threshold Selection</b> | <b>Yes (Centralized Logistic Regression with identical pipeline)</b> |

Table1 — Comparison of related federated learning studies on disease prediction

### III. METHODOLOGY

The proposed system uses a controlled framework to compare logistic regression for diabetes prediction across centralized and federated settings. The centralized approach applies the algorithm to preprocessed and class-imbalance

corrected data in order to obtain the baseline model. The federated approach implements FedProx training of the same model on a per-client basis and aggregates the updates on the server side.

Apart from the baseline comparison, the federated experiments include variations of client distributions as well

as the size of federated systems (number of participating clients). Centralized and federated approaches are tested under the identical experimental and evaluation setup, so any difference in predictive performance cannot be due to the inconsistency in evaluation. All models use the identical decision threshold that maximizes F1-score on out-of-fold validation and are evaluated on the identical test dataset using accuracy, precision, recall, F1-score, ROC-AUC, PR-AUC, and confusion matrix. The general architecture of the proposed system is illustrated in Fig. 1.

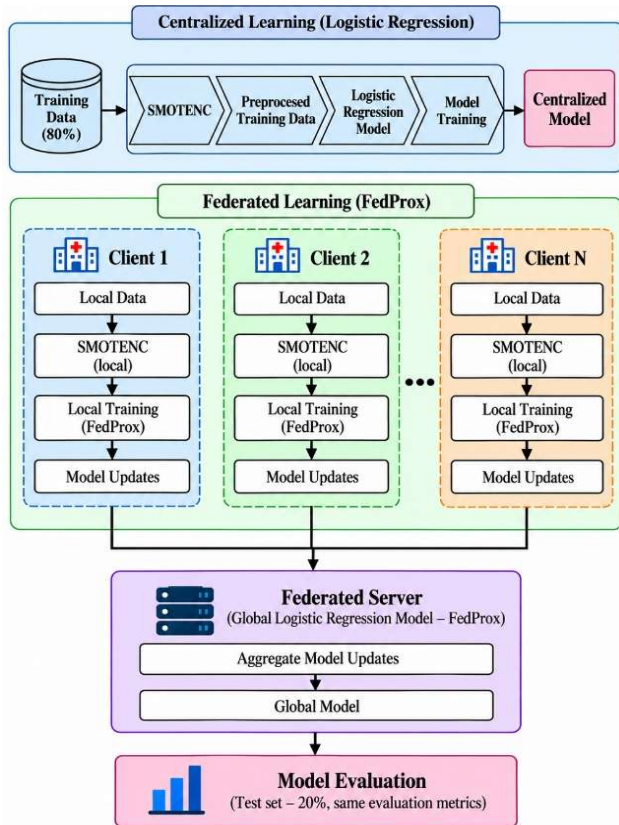


Fig. 1. Architecture Diagram of the System

### A. Dataset and Data Preprocessing

The study uses the Diabetes Prediction Dataset provided by Kaggle, which contains 100,000 instances with 9 attributes having numerical, binary, and categorical variables, where diabetes will be the label.

Before proceeding to model creation, some preliminary cleaning and validation of the dataset need to take place to include deletion of duplicated instances, normalization of categorical text variables, and merging "ever" value in smoking\_history to "former". Some sanity checks on age, bmi, HbA1c\_level, and blood\_glucose\_level variables for invalid data, as well as checking for missing values, will be performed. The cleaned dataset consists of 96,146 records.

This cleaned dataset is now split into two datasets using stratified splitting (random\_state = 42): 80% train dataset and 20% test dataset. All further preprocessing and balancing of the classes will be performed only on the train dataset.

### B. Feature Preprocessing and Class Imbalance Handling

Since the dataset is composed of both numerical, binary, and categorical features, preprocessing is performed independently for each type of features. The numerical

features (age, bmi, HbA1c\_level, and blood\_glucose\_level) are scaled using the StandardScaler. For the binary features (hypertension, heart\_disease) and categorical features (gender, smoking\_history), the OrdinalEncoder method is applied to derive the integer encoding of the data needed by SMOTENC with unknown categories marked with the value of -1. This yields an intermediate eight-dimensional dataset representation where features 0-3 are continuous while features 4-7 are discrete.

SMOTENC takes care of the class imbalance problem in the training dataset because the dataset features include both numerical and categorical data with categorical feature indices being [4, 5, 6, 7]. In the centralized setting, SMOTENC is applied on the whole training dataset (and per fold in cross-validation). In the federated setting, SMOTENC operates independently on each client and synthetic examples do not become shared or aggregated like in prior client-side oversampling approaches [13]. Clients with less than six minority examples do not perform local oversampling to prevent degenerate nearest neighbor search.

After oversampling, categorical and binary features are converted using one-hot encoding (drop="first") while the standardized numerical features are retained. This final feature representation feeds the logistic regression model. The untouched test set never undergoes SMOTENC or any other resampling.

### C. Centralized Learning Model

A centralized logistic regression model serves as the baseline for comparison with the federated approach. It is trained on the preprocessed, SMOTENC-balanced training data, with max\_iter=1000 and random\_state=42.

To have a common classification threshold for the experiments, a stratified three-fold-out-of-fold (OOF) procedure is applied to the run centralized training data. In the process, preprocessing, SMOTENC, model training and probability computation take place using training portion of the data and only probabilities are computed for validation portion. Taking the outputs of three folds together gives us the OOF probabilities of all training samples.

Candidate thresholds are tested against these OOF probabilities, and the one providing the highest F1-score is chosen as the final threshold value. This is because F1-score provides balanced precision-recall measure when the cost of false positive and false negative errors cannot be determined for the particular project. In addition, since each prediction is made by a model trained without seeing the concerned instance, we get the threshold value without any effect of reserved test dataset at all.

The OOF procedure settled on a threshold of 0.8752 (0.8751839224 before rounding). Using a single threshold across every model matters here: if each model's threshold were tuned separately, differences in precision and recall could just reflect where each model's operating point happened to land, not real differences in the underlying model. Fixing the threshold allows differences between the centralized, IID, and non-IID models to be evaluated under the same decision rule. Because the threshold is tuned on the centralized model, it may not suit each federated model's probability calibration equally well; the 0.50 results in Table 6 give a second reference point.

The standard 0.50 threshold is also evaluated as a reference point, while the OOF-selected threshold remains the primary decision rule. Table 6 compares the centralized, IID, and non-IID models at 0.50. Results at the OOF-selected threshold appear in Tables 2, 5, and 7.

#### D. Federated Learning Module

The federated learning implementation uses logistic regression with FedProx-based local optimization. Unlike the centralized implementation, where parameter optimization is handled by `sklearn.LogisticRegression`, the federated implementation performs the logistic regression optimization directly so the FedProx proximal term can be incorporated into the local gradient update.

The logistic regression model computes the linear output and predicted probability as

$$z = Xw + b \quad (1)$$

$$\hat{y} = \sigma(z) = \frac{1}{1+e^{-z}} \quad (2)$$

where  $X$  represents the input features,  $w$  the model weights, and  $b$  the bias. During local training, the weight gradient is calculated using (3):

$$\nabla_w = \frac{1}{m} X^T (\hat{y} - y) + \lambda w + \mu (w - w_{global}) \quad (3)$$

where  $m$  is the mini-batch size,  $y$  is the true label,  $\lambda$  is the L2 regularization coefficient, and  $\mu$  is the FedProx coefficient. The final term in (3) penalizes deviation of the local parameters from the current global parameters.

The federated process follows a server-client architecture. The server initializes the global parameters before the first round and sends the current global parameters to the clients at the start of every communication round. Each client starts from these parameters and trains for 3 local epochs on mini-batches of 128 samples. Before local training, SMOTENC is applied independently at each client to address class imbalance, and the synthetic samples stay within that client and are never sent to the server. After local training, each client returns its updated parameters, and the server aggregates them into the next global model using sample-size-weighted averaging.

Unless otherwise specified, the federated experiments use 3 local epochs, a batch size of 128, a learning rate of 0.10, L2 regularization of  $1 \times 10^{-4}$ , a FedProx coefficient of  $\mu=0.05$ , and 40 communication rounds.

#### E. Non-IID Federated Learning

For the main non-IID experiment, training data is distributed across three clients using class-wise Dirichlet

partitioning, with diabetic and non-diabetic samples partitioned separately. For each class, client allocation proportions are sampled as (4):

$$p \sim \text{Dirichlet}(\alpha, \dots, \alpha) \quad (4)$$

These proportions determine how each class's samples are distributed among clients, after which the class-specific allocations are combined to form each client's dataset. A smaller  $\alpha$  produces greater skew in the allocation, while larger values result in more even distributions; the selected  $\alpha=0.30$  creates heterogeneous class distributions across clients by design.

This setting enables the impact of statistical heterogeneity at the level of clients to be studied without changing the predictive model, pre-processing approach, federated optimization algorithm, and evaluation methodology..

#### F. IID Federated Learning

The IID experiments distribute the training dataset among three clients through stratified distribution, thus ensuring that each client gets a class distribution equal to that of the entire training dataset. Unlike in the non-IID setup, there is no need for Dirichlet distribution or the use of  $\alpha$  in this case.

The FedProx federated learning methodology used in the non-IID experiments is used in the current IID experiments, with the only difference being the distribution of client data. This allows a comparison of the two setups.

#### G. Client-Count Investigation

To examine the impact of federation size, the number of involved clients is altered between 3, 5, and 10 for both IID and non-IID scenarios. The IID scenario uses stratified sampling, while the non-IID scenario uses class-wise Dirichlet sampling with  $\alpha=0.30$ . All of these cases use the same training settings from Section III-D, with 40 communication rounds, and the evaluation threshold and process are also fixed, meaning the optimization process remains constant, while the number of clients varies.

The non-IID client-count experiment should be taken with a grain of salt. The distribution of the clients varies greatly across the different number of clients. Altering the number of clients means altering the Dirichlet partition, the composition of the clients, and their sample sizes. The experiment thus fails to isolate the impact of the number of clients from other related factors.

## IV. RESULTS AND ANALYSIS

In this section, the experimental findings for both the centralized model and the federated model are presented along with the influence of client data distribution on the federated model. The centralized model sets the reference performance, and the three-client non-IID experiment is compared against it under the same evaluation setting. Results for the IID setting, the conventional 0.50 threshold, and 3, 5, and 10 clients follow, along with the classification counts behind the non-IID results.

| Model            | Threshold | Accuracy | Precision | Recall | F1-Score | ROC-AUC | PR-AUC |
|------------------|-----------|----------|-----------|--------|----------|---------|--------|
| Centralized (LR) | 0.8752    | 95.51%   | 78.70%    | 67.33% | 72.58%   | 0.9600  | 0.8170 |

Table2 — Centralized Logistic Regression performance

The centralized logistic regression model, evaluated using the common threshold of 0.8752, achieved an accuracy of 95.51%, precision of 78.70%, recall of 67.33%, and F1-score of 72.58%, with a ROC-AUC of 0.9600 and PR-AUC of 0.8170, as shown in Table 2. The high ROC-AUC indicates strong discrimination between diabetic and non-diabetic cases, while the precision, recall, and F1-score reflect the model's performance at the selected operating threshold. These results establish the centralized model as the reference baseline for evaluating the corresponding federated configurations.

| Client   | Samples | Diabetic | Diabetic proportion |
|----------|---------|----------|---------------------|
| Client 1 | 62,357  | 2,495    | 4.00%               |
| Client 2 | 4,765   | 4,282    | 89.86%              |
| Client 3 | 9,794   | 9        | 0.09%               |

Table3 — Non-IID client distributions

For the non-IID experiment, the three clients held diabetic-class proportions of 4.00%, 89.86%, and 0.09%, so the label distributions are strongly skewed across clients. Table 3 lists the client sizes and class counts. This three-client configuration is the base federated setting for comparison with the centralized model. Despite the skew, the federated model

reached 95.65% accuracy, 80.31% precision, 67.10% recall, and a 73.11% F1-score, with a ROC-AUC of 0.9589 and a PR-AUC of 0.8150 (Table 5). These results are close to the centralized baseline of 95.51% accuracy, 78.70% precision, 67.33% recall, and 72.58% F1-score. In this tested configuration, federated training therefore kept comparable predictive performance without sending clients' raw training data to the server.

| Client   | Samples | Diabetic | Diabetic Proportion |
|----------|---------|----------|---------------------|
| Client 1 | 25,639  | 2,262    | 8.82%               |
| Client 2 | 25,639  | 2,262    | 8.82%               |
| Client 3 | 25,638  | 2,262    | 8.82%               |

Table4 — IID Client-Level Class Distribution

Under the IID setting, the three clients maintained approximately the same diabetic-class proportion of 8.82% (Table 4), representing a comparatively homogeneous client distribution. The resulting federated model achieved 95.49% accuracy, 78.55% precision, 67.16% recall, and 72.41% F1-score, with a ROC-AUC of 0.9599 and PR-AUC of 0.8168 (Table 5). These values were also close to the centralized baseline, indicating that the three-client federated model maintained comparable predictive performance when client class distributions were closely aligned.

| Configuration | Accuracy | Precision | Recall | F1-score | ROC-AUC | PR-AUC |
|---------------|----------|-----------|--------|----------|---------|--------|
| IID           | 95.49%   | 78.55%    | 67.16% | 72.41%   | 0.9599  | 0.8168 |
| Non-IID       | 95.65%   | 80.31%    | 67.10% | 73.11%   | 0.9589  | 0.8150 |

Table5 — Federated Learning Performance (3 Clients)

| Model       | Clients | Accuracy | Precision | Recall | F1-score | ROC-AUC | PR-AUC |
|-------------|---------|----------|-----------|--------|----------|---------|--------|
| Centralized | —       | 88.43%   | 42.47%    | 87.85% | 57.26%   | 0.9600  | 0.8170 |
| IID         | 3       | 88.32%   | 42.20%    | 87.74% | 56.99%   | 0.9599  | 0.8168 |
| IID         | 5       | 88.38%   | 42.34%    | 87.62% | 57.09%   | 0.9598  | 0.8170 |
| IID         | 10      | 88.52%   | 42.62%    | 87.21% | 57.26%   | 0.9594  | 0.8158 |
| Non-IID     | 3       | 88.71%   | 43.02%    | 86.56% | 57.48%   | 0.9589  | 0.8150 |
| Non-IID     | 5       | 89.50%   | 44.96%    | 85.14% | 58.84%   | 0.9563  | 0.8017 |
| Non-IID     | 10      | 94.76%   | 70.04%    | 70.99% | 70.51%   | 0.9567  | 0.8074 |

Table6 — 0.5 Threshold Results

Table6 reports the centralized and federated models at the standard 0.50 threshold. At this operating point, most models reach high recall but low precision. The centralized model has a recall of 87.85%, so it catches most diabetic cases, but its precision is only 42.47%, meaning that many of the cases it flags as diabetic are false positives. Most federated configurations show the same pattern, with precision between 42.20% and 70.04% and recall between 70.99% and 87.74%. The gain in recall comes with a large increase in false positives, which does not suit the intended evaluation. The 0.50 threshold is therefore kept as a reference point and not as the primary decision threshold.

| Distribution | Clients | Accuracy | Precision | Recall | F1-score | ROC-AUC | PR-AUC |
|--------------|---------|----------|-----------|--------|----------|---------|--------|
| IID          | 3       | 95.49%   | 78.55%    | 67.16% | 72.41%   | 0.9599  | 0.8168 |
| IID          | 5       | 95.50%   | 78.70%    | 67.10% | 72.44%   | 0.9598  | 0.8170 |

|         |    |        |        |        |        |        |        |
|---------|----|--------|--------|--------|--------|--------|--------|
| IID     | 10 | 95.51% | 78.65% | 67.33% | 72.55% | 0.9594 | 0.8158 |
| Non-IID | 3  | 95.65% | 80.31% | 67.10% | 73.11% | 0.9589 | 0.8150 |
| Non-IID | 5  | 95.14% | 74.74% | 67.87% | 71.14% | 0.9563 | 0.8017 |
| Non-IID | 10 | 95.35% | 97.51% | 48.53% | 64.80% | 0.9567 | 0.8074 |

Table7 — Federated Performance with Different Client Counts

Table 7 reports federated performance with 3, 5, and 10 clients under IID and non-IID partitioning.

Under IID partitioning, adding clients barely changed the results. Accuracy stayed between 95.49% and 95.51%, and precision, recall, F1-score, ROC-AUC, and PR-AUC shifted only slightly. This is consistent with the roughly equal class proportions across IID clients: within the tested range, the number of clients had little effect on aggregate performance.

The non-IID results varied more, and the client distributions behind them are uneven. As Fig. 2 shows, diabetic-class proportions range from 0% to 95.63% across the ten clients, and several clients contain no diabetic samples. With three clients, the model reached 95.65% accuracy, 80.31% precision, 67.10% recall, and 73.11% F1-score. At five clients, precision dropped to 74.74% and F1-score to 71.14%, while recall edged up to 67.87%. At ten clients, precision jumped to 97.51% but recall fell to 48.53%, giving an F1-score of 64.80%. Fig. 3 shows this shift most clearly at

ten clients. At the common threshold, that model is more conservative: it makes fewer false-positive predictions but misses more diabetic cases.

The client distribution is not the whole explanation. For the same ten-client non-IID configuration, the 0.50 threshold gave 70.04% precision and 70.99% recall, while the common threshold of 0.8752 gave 97.51% precision and 48.53% recall. The higher threshold therefore accounts for a large part of the precision gain and recall loss, alongside the uneven class distributions across clients.

Those distributions are very uneven. Fig. 3 shows diabetic-class proportions from 0% to 95.63% across the ten clients, and several clients contain no diabetic samples. Changing the number of clients also changes the Dirichlet partition, the client composition, and the client sample sizes, so these experiments cannot separate the effect of client count from those changes. The results describe how these particular generated configurations behaved. They do not show that client count alone causes the differences.

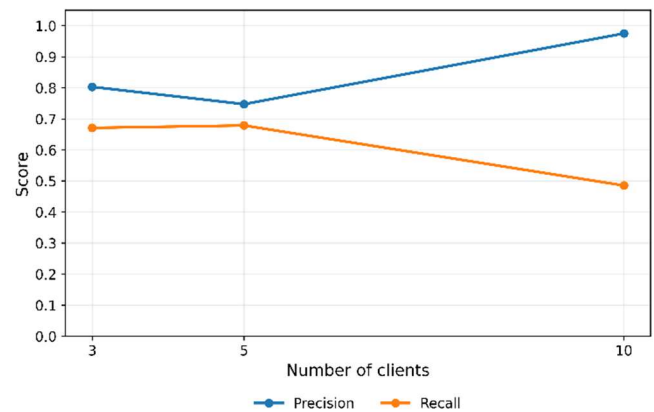
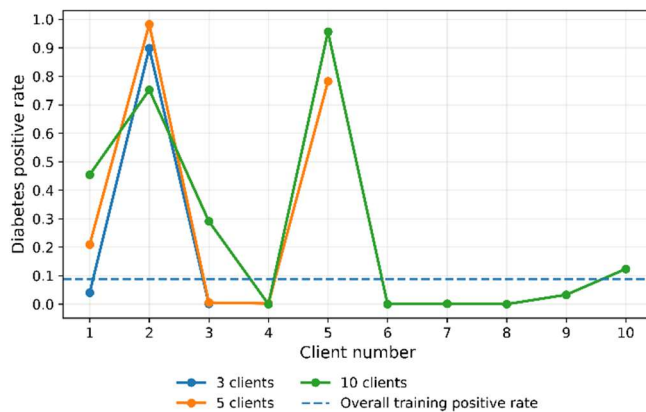


Fig. 2. Non-IID client class distributions Fig. 3. Precision-recall trend across Non-IID client counts

| Non-IID Clients | TP    | FP  | TN     | FN  |
|-----------------|-------|-----|--------|-----|
| 3               | 1,138 | 279 | 17,255 | 558 |
| 5               | 1,151 | 389 | 17,145 | 545 |
| 10              | 823   | 21  | 17,513 | 873 |

Table8 —Non-IID Classification Counts

Table8 gives the classification counts behind the non-IID results. At the common threshold, the ten-client model produces only 21 false positives, against 279 for three clients and 389 for five. That drop accounts for its precision of 97.51%. The same model also produces 873 false negatives, against 558 and 545, so it finds fewer diabetic cases and its recall falls to 48.53%. The high precision of the ten-client model therefore comes with many more missed diabetic cases. It is not an improvement across all measures.

## V. CONCLUSION

This paper compared centralized and federated logistic regression for diabetes prediction, with a focus on how client-level statistical heterogeneity affects federated performance. With three clients, the federated model performed comparably to the centralized baseline under both IID and non-IID partitioning, even though the non-IID clients held diabetic-class proportions from 0.09% to 89.86%. Under IID partitioning, performance also stayed stable as the number of clients grew from 3 to 10. Under non-IID partitioning, precision and recall varied more with client count. At ten clients, precision rose to 97.51% while recall fell to 48.53%, and part of that shift comes from the shared decision threshold in addition to the client distributions. Because changing the client count also changes the generated partition, these differences cannot be attributed to client count alone. Overall, the results support federated

learning as a way to build a diabetes prediction model across institutions without sending clients' raw training data to a central server. The clients here were simulated by partitioning a single dataset, so these findings still need to be checked on real multi-institution data.

Future research in this project would expand into multiple directions as follows: use of differential privacy for better data privacy, application of multi-task federated learning for multi-disease diagnosis and development of federated optimization that can be used in a heterogeneous client setting. Another important aspect to consider in the future would be a dashboard for overall evaluation of the system.

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# Federated Learning for Diabetes Prediction: Investigating Client-Level Statistical Heterogeneity

**Abstract**—Diabetes prediction using machine learning can support early identification of individuals at risk, however, existing centralized methods rely on patient data being collected and aggregated into a central repository. This raises data sharing and data governance issues for healthcare organizations that must cooperate for this purpose. Another approach is federated learning in which a joint model is trained cooperatively by healthcare organizations and raw data does not leave the respective organization. Although the performance of federated learning depends on the degree of class imbalance, differences in data distributions on the side of the clients, as well as the number of clients, its exact effect is yet unknown. This research explores the above factors via a comparative study of centralized and federated learning for predicting diabetes. Logistic regression is the base prediction model used, while class imbalance is solved by SMOTENC algorithm. In federated learning, local SMOTENC technique is performed separately for each client, local training is performed using FedProx, and server aggregation is performed using sample size weighted average aggregation technique. Evaluation is performed for both IID and non-IID client data distributions, where non-IID client data distributions are performed using Dirichlet distribution ( $\alpha = 0.30$ ). The study also examines the impact of the change of the number of clients from 3 to 5 and 10. Evaluation of the resulting models is done on the same unseen testing set, using the accuracy, precision, recall, F1 score, ROC-AUC, and PR-AUC metrics. The outcomes allow comparing the performance of centralized and federated approaches to diabetes prediction.

**Keywords**—federated learning, diabetes prediction, logistic regression, class imbalance, SMOTENC, FedProx, client heterogeneity

## I. INTRODUCTION

Diabetes mellitus is a chronic disease that affected an estimated 828 million adults worldwide in 2022 [11] and can lead to serious complications such as cardiovascular disease, renal failure, and diabetic retinopathy. Early and accurate prediction of diabetes is therefore of considerable importance, since it can support timely intervention, better disease management, and healthcare planning [1]. With the growing availability and use of healthcare data, machine learning techniques are increasingly applied to identify patterns in clinical information and support disease prediction and decision-making. The growth in medical data also brings challenges in computation, security, governance, and cross-institutional sharing, particularly when information comes from multiple healthcare institutions [4].

The use of centralized machine learning requires patient records from different institutions to be collected in a common repository before training. Although this approach allows models to learn from a larger and more diverse dataset, healthcare data is highly sensitive and subject to restrictions related to privacy, governance, so centralizing it creates security risks. Healthcare institutions may therefore be unable or unwilling to share raw patient records [2], creating a significant challenge for machine learning which benefits from diverse data yet patient information is heavily

governed by institutional regulations meaning data cannot be seamlessly transferred between organizations.

Federated learning provides a distributed solution to this issue by enabling several healthcare organizations to jointly learn a model without sharing any data at all. Instead of gathering data into one place, the participants will be training models locally and will only share the output, i.e., the updates to the model [3]. Such a method has attracted much interest in medicine since it enables the organization to learn jointly while keeping their patients' information secure. Thus, federated learning has been used for some clinical applications, such as diabetes prediction and electronic health record analysis [6].

Related diabetes prediction studies have used techniques such as the Synthetic Minority Over-sampling Technique (SMOTE) to address class imbalance and have compared different machine learning and federated learning approaches [1]. Client-level oversampling combined with FedProx has also been applied to imbalanced cardiovascular risk data, where FedProx outperformed standard FedAvg [13]. In practice, client data is often non-IID: class proportions and feature distributions differ across institutions, which can affect convergence and accuracy in federated training [12]. Federated learning research in healthcare has also examined distributed prediction with different numbers of clients, showing that data distribution and system configuration affect model performance [3]. Although further investigation is required in understanding how centralized learning and federated learning compare when the same prediction methodology is maintained, few studies have jointly considered factors such as class imbalance, IID and non-IID data distributions, and changes in the number of participating clients.

This study therefore investigates how client-level statistical heterogeneity affects federated learning performance for diabetes prediction and compares the resulting federated model with an equivalent centralized model. More precisely, it examines how heterogeneity in client data affects predictive performance relative to centralized learning, and how the gap between the two differs between IID and non-IID client distributions. It also evaluates 3, 5, and 10 clients under both settings to see how federation size and data distribution affect performance. Logistic regression is the prediction model, SMOTENC (Synthetic Minority Over-sampling Technique for Nominal and Continuous features) handles class imbalance, and FedProx is used for federated training. A single classification threshold is selected once, using three-fold out-of-fold predictions from the centralized training data to maximize F1-score, and is then fixed across all centralized and federated experiments. Performance is evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC.

## II. LITERATURE SURVEY

Dash et al. [1] study privacy-preserving methods of predicting diabetes and heart diseases based on federated learning due to the necessity of early diabetes diagnosis and difficulties with data exchange in the health-care sector in the context of centralized machine learning. FL with Weighted Conglomeration Optimization (FLWCO) method by the authors is aimed at selecting local classifiers according to their performance level and using confidence score thresholds, which improves communication efficiency and convergence rate.

Grant et al. [2] compare federated learning with centralized learning for diabetic classification with respect to the PIMA Indians Diabetes Dataset, where they apply logistic regression to determine if federated learning is able to equal centralized learning without requiring central collection of the raw data. Federated learning achieves ROC-AUC score of 0.77, which is close to the baseline of 0.78 achieved with centralized learning, making their paper an important reference point in comparing these methods with structured diabetes data. However, the authors do not consider the problem of non-IID data distribution.

Haripriya et al. [3] present a Federated Learning approach with adaptive aggregation for privacy-enabled image classification in medical images. With the use of chest X-rays, brain MRIs, and diabetic retinopathy datasets, the dynamic aggregation method of Haripriya et al., which dynamically changes between FedAvg and FedSGD according to the weight variation of the clients, achieves up to 98.6% accuracy in classifying brain tumors with fine-tuning on EfficientNetV2. This work mainly evaluates its approach through a simulated system consisting of ten clients. Initial tests with up to 100 clients reveal increased communication cost and reduced accuracy, raising the issue of scalability to hundreds of clients.

Demir et al. [4] study privacy-preserving diabetes and heart disease prediction with federated learning, motivated by the privacy, security, and regulatory limits on centralizing medical records. Their Data Unaware Classification Based on Association (duCBA) algorithm aggregates logical relational rules instead of high-dimensional model weights, which improves privacy and computational efficiency while keeping decisions interpretable. Statistical heterogeneity across clients is not addressed.

Atmoko et al. [5] propose FedXChain, an explainable federated learning framework that addresses heterogeneous and non-IID healthcare data through SHAP-based aggregation, adaptive trust-weighted aggregation, and blockchain auditing. On the Wisconsin Breast Cancer dataset, their logistic regression model reaches 96.5% accuracy. The evaluation uses a fixed ten-client setup and compares only against federated baselines (FedAvg and FedProx), so the effect of client count, and any comparison with an equivalent centralized model, is left unexamined.

Across these studies, the federated learning literature for disease prediction has concentrated on developing better aggregation, privacy, or optimization mechanisms, most often under a fixed client configuration. Most include a centralized baseline, but few examine how the gap between centralized and federated performance changes with client data distribution. Among the studies that scale the number of clients, Haripriya et al. and Shahid et al. evaluate non-IID partitions only, and Demir et al. evaluates IID partitions only. No reviewed study compares IID and non-IID partitions across client counts against a centralized baseline while also handling class imbalance. This leaves open how class imbalance, IID versus non-IID client

Bebortta et al. [6] examine privacy preserving prediction of cardiac risks using federated learning on IoT-enabled Electronic Health Records (EHR). The proposed FedEHR approach employs a sparse support vector machine, a cluster primal-dual splitting algorithm, and round-robin heuristic to enhance the predictive ability while cutting down on communication and computation overheads. The paper recognizes data quality, heterogeneity, and scalability as open problems, but it does not analyze the influence of different client-side data distribution on the performance disparity of centralized versus federated learning approaches.

Shahid et al. [7] examine privacy-preserving chronic disease prediction using federated learning, motivated by the need for early diagnosis and the privacy, regulatory, and interoperability challenges of centralizing patient data. Their framework, Health-FedNet, combines calibrated differential privacy, Paillier homomorphic encryption, and adaptive node weighting, improving predictive accuracy, cutting communication overhead by 41.6%, and lowering data-leakage risk.

Rahman et al. [8] target privacy-preserving early stroke risk prediction and continuous patient monitoring through a federated, IoT-enabled machine learning framework. Their approach pairs a Federated Averaging (FedAvg) pipeline with a wearable sensing system, combining multi-stage feature optimization (PSO, mRMR, RFC, and PCA/LDA) with real-time limb-temperature asymmetry and blood-oxygen (SaO<sub>2</sub>) tracking, reaching up to 99.44% accuracy. Because the evaluation relies on a single Kaggle dataset with stratified IID partitions, validation under non-IID client variation is left for future work.

Sorayaie Azar et al. [9] investigate privacy-preserving coronary artery disease prediction using federated learning. Their framework, FMLCA, combines decentralized machine learning classifiers with k-anonymity, implemented via the IBM AI Privacy Toolkit, and SHAP-based explainability to protect patient confidentiality and support clinical trust, reaching 83.21% classification accuracy. The evaluation is limited to a relatively small clinical dataset, so performance across diverse, heterogeneous multi-center settings remains untested.

Amin et al. [10] propose Distribution-Aware Federated Learning (DA-FL) for diabetes prediction on the CDC BRFSS 2021 dataset, addressing non-IID data and class imbalance through class-weighted local training and distribution-aware aggregation. The work centers on the aggregation strategy itself and uses a fixed five-client federation, leaving the effect of varying client numbers, and the resulting performance shifts under IID and non-IID conditions, less explored.

Table 1 summarizes these studies alongside the present work.

distributions, and the number of participating clients jointly affect the gap between centralized and federated performance, which is the focus of the present work.

| Work                 | Dataset                                                                    | FL Method                                                            | IID / Non-IID                                                        | Client Count                     | Imbalance Handling                                                                  | Centralized Baseline                                                 |
|----------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| [1]                  | Kaggle Diabetes (96,146 records)                                           | Weighted Conglomeration Optimization (FLWCO), with FedAvg/FedMAP     | Not specified                                                        | 4 clients                        | SMOTE                                                                               | <b>Yes</b> (Centralized RF, XGBoost, MLP, Gradient Boost, AdaBoost)  |
| [2]                  | PIMA                                                                       | Flower federated Logistic Regression                                 | IID                                                                  | 3 clients                        | None (Evaluated natural class skew)                                                 | <b>Yes</b> (Centralized RF, SVM, LR)                                 |
| [3]                  | Multimodal Medical Images                                                  | Adaptive Aggregation (FedAvg / FedSGD switching) + Transfer Learning | Non-IID (random vs. stratified)                                      | 10 clients (Evaluated up to 100) | None (Pretrained ImageNet weights)                                                  | <b>Yes</b> (Centralized Transfer Learning with DP)                   |
| [4]                  | CDC Diabetes Health Indicators and the Kaggle Heart Attack dataset (Fayez) | duCBA (Data-Unaware Classification Based on Association) vs. FedAvg  | IID                                                                  | 2, 4, 8 clients                  | Random under-sampling on Heart Attack; the diabetes set is already balanced.        | <b>Yes</b> (Centralized CBA)                                         |
| [5]                  | Wisconsin Breast Cancer (WDBC) & Synthetic set                             | FedXChain (Adaptive Trust Scoring + Blockchain + SHAP)               | Both IID & Non-IID (Dirichlet $\alpha = 0.3, 0.5$ )                  | 10 nodes                         | Not specified                                                                       | <b>No</b>                                                            |
| [6]                  | de-identified EHRs from Boston Medical Center                              | FedEHR (Sparse SVM + CPDS + Round-Robin Heuristic)                   | Not specified                                                        | 5 clients                        | Not specified                                                                       | <b>Yes</b> (Centralized SVM)                                         |
| [7]                  | MIMIC-III, diabetes and hypertension                                       | Health-FedNet (Adaptive Node Weighting + DP + Paillier HE)           | Non-IID                                                              | 5 to 100 clients                 | Threshold tuning / None                                                             | <b>Yes</b> (Centralized Deep Learning)                               |
| [8]                  | Kaggle Stroke Risk Prediction Dataset Based on Symptoms (70,000 records)   | FedAvg + Multi-stage Feature Selection (PSO, mRMR, RFC)              | IID (Stratified; Non-IID left as future work)                        | 10 clients (Fixed)               | None (Evaluated via F1 / MCC)                                                       | <b>Yes</b> (Centralized XGBoost, RF, MLP)                            |
| [9]                  | UCI Heart Disease (Cleveland), 303 samples                                 | FMLCA (Federated ML + k-anonymity + SHAP)                            | Not specified                                                        | 5 clients                        | SMOTE                                                                               | <b>No</b>                                                            |
| [10]                 | CDC BRFS 2021 Diabetes (236,378 records)                                   | DA-FL (Distribution-Aware FL + Minority Amplification $\phi_k$ )     | Non-IID (Dirichlet $\alpha \in [0.1, 1.0]$ )                         | 5 clients (Fixed)                | Class-Weighted Cross-Entropy + Minority Amplification                               | <b>Yes</b> (Centralized MLP)                                         |
| <b>Proposed Work</b> | <b>Kaggle Diabetes Prediction Dataset (96,146 records)</b>                 | <b>FedProx (Federated Logistic Regression with Proximal Term)</b>    | <b>Both IID &amp; Non-IID (Dirichlet <math>\alpha = 0.30</math>)</b> | <b>Varied (3, 5, 10 clients)</b> | <b>Local SMOTENC per client + Centralized OOF F1-Maximizing Threshold Selection</b> | <b>Yes (Centralized Logistic Regression with identical pipeline)</b> |

Table1 — Comparison of related federated learning studies on disease prediction

### III. METHODOLOGY

The proposed system uses a controlled framework to compare logistic regression for diabetes prediction across centralized and federated settings. The centralized approach applies the algorithm to preprocessed and class-imbalance

corrected data in order to obtain the baseline model. The federated approach implements FedProx training of the same model on a per-client basis and aggregates the updates on the server side.

Apart from the baseline comparison, the federated experiments include variations of client distributions as well

as the size of federated systems (number of participating clients). Centralized and federated approaches are tested under the identical experimental and evaluation setup, so any difference in predictive performance cannot be due to the inconsistency in evaluation. All models use the identical decision threshold that maximizes F1-score on out-of-fold validation and are evaluated on the identical test dataset using accuracy, precision, recall, F1-score, ROC-AUC, PR-AUC, and confusion matrix. The general architecture of the proposed system is illustrated in Fig. 1.

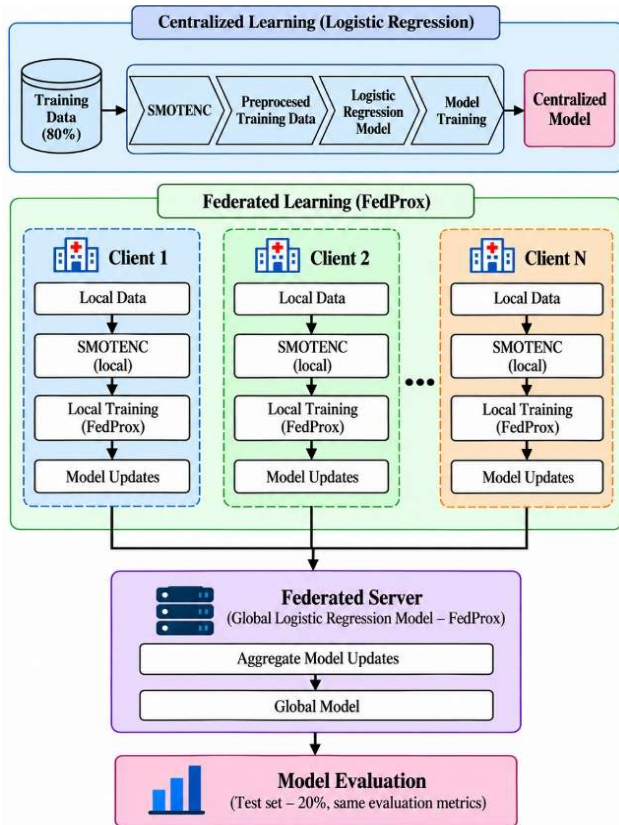


Fig. 1. Architecture Diagram of the System

#### A. Dataset and Data Preprocessing

The study uses the Diabetes Prediction Dataset provided by Kaggle, which contains 100,000 instances with 9 attributes having numerical, binary, and categorical variables, where diabetes will be the label.

Before proceeding to model creation, some preliminary cleaning and validation of the dataset need to take place to include deletion of duplicated instances, normalization of categorical text variables, and merging "ever" value in smoking\_history to "former". Some sanity checks on age, bmi, HbA1c\_level, and blood\_glucose\_level variables for invalid data, as well as checking for missing values, will be performed. The cleaned dataset consists of 96,146 records.

This cleaned dataset is now split into two datasets using stratified splitting (random\_state = 42): 80% train dataset and 20% test dataset. All further preprocessing and balancing of the classes will be performed only on the train dataset.

#### B. Feature Preprocessing and Class Imbalance Handling

Since the dataset is composed of both numerical, binary, and categorical features, preprocessing is performed independently for each type of features. The numerical

features (age, bmi, HbA1c\_level, and blood\_glucose\_level) are scaled using the StandardScaler. For the binary features (hypertension, heart\_disease) and categorical features (gender, smoking\_history), the OrdinalEncoder method is applied to derive the integer encoding of the data needed by SMOTENC with unknown categories marked with the value of -1. This yields an intermediate eight-dimensional dataset representation where features 0-3 are continuous while features 4-7 are discrete.

SMOTENC takes care of the class imbalance problem in the training dataset because the dataset features include both numerical and categorical data with categorical feature indices being [4, 5, 6, 7]. In the centralized setting, SMOTENC is applied on the whole training dataset (and per fold in cross-validation). In the federated setting, SMOTENC operates independently on each client and synthetic examples do not become shared or aggregated like in prior client-side oversampling approaches [13]. Clients with less than six minority examples do not perform local oversampling to prevent degenerate nearest neighbor search.

After oversampling, categorical and binary features are converted using one-hot encoding (drop="first") while the standardized numerical features are retained. This final feature representation feeds the logistic regression model. The untouched test set never undergoes SMOTENC or any other resampling.

#### C. Centralized Learning Model

A centralized logistic regression model serves as the baseline for comparison with the federated approach. It is trained on the preprocessed, SMOTENC-balanced training data, with max\_iter=1000 and random\_state=42.

To have a common classification threshold for the experiments, a stratified three-fold-out-of-fold (OOF) procedure is applied to the run centralized training data. In the process, preprocessing, SMOTENC, model training and probability computation take place using training portion of the data and only probabilities are computed for validation portion. Taking the outputs of three folds together gives us the OOF probabilities of all training samples.

Candidate thresholds are tested against these OOF probabilities, and the one providing the highest F1-score is chosen as the final threshold value. This is because F1-score provides balanced precision-recall measure when the cost of false positive and false negative errors cannot be determined for the particular project. In addition, since each prediction is made by a model trained without seeing the concerned instance, we get the threshold value without any effect of reserved test dataset at all.

The OOF procedure settled on a threshold of 0.8752 (0.8751839224 before rounding). Using a single threshold across every model matters here: if each model's threshold were tuned separately, differences in precision and recall could just reflect where each model's operating point happened to land, not real differences in the underlying model. Fixing the threshold allows differences between the centralized, IID, and non-IID models to be evaluated under the same decision rule. Because the threshold is tuned on the centralized model, it may not suit each federated model's probability calibration equally well; the 0.50 results in Table 6 give a second reference point.

The standard 0.50 threshold is also evaluated as a reference point, while the OOF-selected threshold remains the primary decision rule. Table 6 compares the centralized, IID, and non-IID models at 0.50. Results at the OOF-selected threshold appear in Tables 2, 5, and 7.

#### D. Federated Learning Module

The federated learning implementation uses logistic regression with FedProx-based local optimization. Unlike the centralized implementation, where parameter optimization is handled by `sklearn.LogisticRegression`, the federated implementation performs the logistic regression optimization directly so the FedProx proximal term can be incorporated into the local gradient update.

The logistic regression model computes the linear output and predicted probability as

$$z = Xw + b \quad (1)$$

$$\hat{y} = \sigma(z) = \frac{1}{1+e^{-z}} \quad (2)$$

where  $X$  represents the input features,  $w$  the model weights, and  $b$  the bias. During local training, the weight gradient is calculated using (3):

$$\nabla_w = \frac{1}{m} X^T (\hat{y} - y) + \lambda w + \mu (w - w_{global}) \quad (3)$$

where  $m$  is the mini-batch size,  $y$  is the true label,  $\lambda$  is the L2 regularization coefficient, and  $\mu$  is the FedProx coefficient. The final term in (3) penalizes deviation of the local parameters from the current global parameters.

The federated process follows a server-client architecture. The server initializes the global parameters before the first round and sends the current global parameters to the clients at the start of every communication round. Each client starts from these parameters and trains for 3 local epochs on mini-batches of 128 samples. Before local training, SMOTENC is applied independently at each client to address class imbalance, and the synthetic samples stay within that client and are never sent to the server. After local training, each client returns its updated parameters, and the server aggregates them into the next global model using sample-size-weighted averaging.

Unless otherwise specified, the federated experiments use 3 local epochs, a batch size of 128, a learning rate of 0.10, L2 regularization of  $1 \times 10^{-4}$ , a FedProx coefficient of  $\mu=0.05$ , and 40 communication rounds.

#### E. Non-IID Federated Learning

For the main non-IID experiment, training data is distributed across three clients using class-wise Dirichlet

partitioning, with diabetic and non-diabetic samples partitioned separately. For each class, client allocation proportions are sampled as (4):

$$p \sim \text{Dirichlet}(\alpha, \dots, \alpha) \quad (4)$$

These proportions determine how each class's samples are distributed among clients, after which the class-specific allocations are combined to form each client's dataset. A smaller  $\alpha$  produces greater skew in the allocation, while larger values result in more even distributions; the selected  $\alpha=0.30$  creates heterogeneous class distributions across clients by design.

This setting enables the impact of statistical heterogeneity at the level of clients to be studied without changing the predictive model, pre-processing approach, federated optimization algorithm, and evaluation methodology..

#### F. IID Federated Learning

The IID experiments distribute the training dataset among three clients through stratified distribution, thus ensuring that each client gets a class distribution equal to that of the entire training dataset. Unlike in the non-IID setup, there is no need for Dirichlet distribution or the use of  $\alpha$  in this case.

The FedProx federated learning methodology used in the non-IID experiments is used in the current IID experiments, with the only difference being the distribution of client data. This allows a comparison of the two setups.

#### G. Client-Count Investigation

To examine the impact of federation size, the number of involved clients is altered between 3, 5, and 10 for both IID and non-IID scenarios. The IID scenario uses stratified sampling, while the non-IID scenario uses class-wise Dirichlet sampling with  $\alpha=0.30$ . All of these cases use the same training settings from Section III-D, with 40 communication rounds, and the evaluation threshold and process are also fixed, meaning the optimization process remains constant, while the number of clients varies.

The non-IID client-count experiment should be taken with a grain of salt. The distribution of the clients varies greatly across the different number of clients. Altering the number of clients means altering the Dirichlet partition, the composition of the clients, and their sample sizes. The experiment thus fails to isolate the impact of the number of clients from other related factors.

## IV. RESULTS AND ANALYSIS

In this section, the experimental findings for both the centralized model and the federated model are presented along with the influence of client data distribution on the federated model. The centralized model sets the reference performance, and the three-client non-IID experiment is compared against it under the same evaluation setting. Results for the IID setting, the conventional 0.50 threshold, and 3, 5, and 10 clients follow, along with the classification counts behind the non-IID results.

| Model            | Threshold | Accuracy | Precision | Recall | F1-Score | ROC-AUC | PR-AUC |
|------------------|-----------|----------|-----------|--------|----------|---------|--------|
| Centralized (LR) | 0.8752    | 95.51%   | 78.70%    | 67.33% | 72.58%   | 0.9600  | 0.8170 |

Table2 — Centralized Logistic Regression performance

The centralized logistic regression model, evaluated using the common threshold of 0.8752, achieved an accuracy of 95.51%, precision of 78.70%, recall of 67.33%, and F1-score of 72.58%, with a ROC-AUC of 0.9600 and PR-AUC of 0.8170, as shown in Table 2. The high ROC-AUC indicates strong discrimination between diabetic and non-diabetic cases, while the precision, recall, and F1-score reflect the model's performance at the selected operating threshold. These results establish the centralized model as the reference baseline for evaluating the corresponding federated configurations.

| Client   | Samples | Diabetic | Diabetic proportion |
|----------|---------|----------|---------------------|
| Client 1 | 62,357  | 2,495    | 4.00%               |
| Client 2 | 4,765   | 4,282    | 89.86%              |
| Client 3 | 9,794   | 9        | 0.09%               |

Table3 — Non-IID client distributions

For the non-IID experiment, the three clients held diabetic-class proportions of 4.00%, 89.86%, and 0.09%, so the label distributions are strongly skewed across clients. Table 3 lists the client sizes and class counts. This three-client configuration is the base federated setting for comparison with the centralized model. Despite the skew, the federated model

reached 95.65% accuracy, 80.31% precision, 67.10% recall, and a 73.11% F1-score, with a ROC-AUC of 0.9589 and a PR-AUC of 0.8150 (Table 5). These results are close to the centralized baseline of 95.51% accuracy, 78.70% precision, 67.33% recall, and 72.58% F1-score. In this tested configuration, federated training therefore kept comparable predictive performance without sending clients' raw training data to the server.

| Client   | Samples | Diabetic | Diabetic Proportion |
|----------|---------|----------|---------------------|
| Client 1 | 25,639  | 2,262    | 8.82%               |
| Client 2 | 25,639  | 2,262    | 8.82%               |
| Client 3 | 25,638  | 2,262    | 8.82%               |

Table4 — IID Client-Level Class Distribution

Under the IID setting, the three clients maintained approximately the same diabetic-class proportion of 8.82% (Table 4), representing a comparatively homogeneous client distribution. The resulting federated model achieved 95.49% accuracy, 78.55% precision, 67.16% recall, and 72.41% F1-score, with a ROC-AUC of 0.9599 and PR-AUC of 0.8168 (Table 5). These values were also close to the centralized baseline, indicating that the three-client federated model maintained comparable predictive performance when client class distributions were closely aligned.

| Configuration | Accuracy | Precision | Recall | F1-score | ROC-AUC | PR-AUC |
|---------------|----------|-----------|--------|----------|---------|--------|
| IID           | 95.49%   | 78.55%    | 67.16% | 72.41%   | 0.9599  | 0.8168 |
| Non-IID       | 95.65%   | 80.31%    | 67.10% | 73.11%   | 0.9589  | 0.8150 |

Table5 — Federated Learning Performance (3 Clients)

| Model       | Clients | Accuracy | Precision | Recall | F1-score | ROC-AUC | PR-AUC |
|-------------|---------|----------|-----------|--------|----------|---------|--------|
| Centralized | —       | 88.43%   | 42.47%    | 87.85% | 57.26%   | 0.9600  | 0.8170 |
| IID         | 3       | 88.32%   | 42.20%    | 87.74% | 56.99%   | 0.9599  | 0.8168 |
| IID         | 5       | 88.38%   | 42.34%    | 87.62% | 57.09%   | 0.9598  | 0.8170 |
| IID         | 10      | 88.52%   | 42.62%    | 87.21% | 57.26%   | 0.9594  | 0.8158 |
| Non-IID     | 3       | 88.71%   | 43.02%    | 86.56% | 57.48%   | 0.9589  | 0.8150 |
| Non-IID     | 5       | 89.50%   | 44.96%    | 85.14% | 58.84%   | 0.9563  | 0.8017 |
| Non-IID     | 10      | 94.76%   | 70.04%    | 70.99% | 70.51%   | 0.9567  | 0.8074 |

Table6 — 0.5 Threshold Results

Table6 reports the centralized and federated models at the standard 0.50 threshold. At this operating point, most models reach high recall but low precision. The centralized model has a recall of 87.85%, so it catches most diabetic cases, but its precision is only 42.47%, meaning that many of the cases it flags as diabetic are false positives. Most federated configurations show the same pattern, with precision between 42.20% and 70.04% and recall between 70.99% and 87.74%. The gain in recall comes with a large increase in false positives, which does not suit the intended evaluation. The 0.50 threshold is therefore kept as a reference point and not as the primary decision threshold.

| Distribution | Clients | Accuracy | Precision | Recall | F1-score | ROC-AUC | PR-AUC |
|--------------|---------|----------|-----------|--------|----------|---------|--------|
| IID          | 3       | 95.49%   | 78.55%    | 67.16% | 72.41%   | 0.9599  | 0.8168 |
| IID          | 5       | 95.50%   | 78.70%    | 67.10% | 72.44%   | 0.9598  | 0.8170 |

|         |    |        |        |        |        |        |        |
|---------|----|--------|--------|--------|--------|--------|--------|
| IID     | 10 | 95.51% | 78.65% | 67.33% | 72.55% | 0.9594 | 0.8158 |
| Non-IID | 3  | 95.65% | 80.31% | 67.10% | 73.11% | 0.9589 | 0.8150 |
| Non-IID | 5  | 95.14% | 74.74% | 67.87% | 71.14% | 0.9563 | 0.8017 |
| Non-IID | 10 | 95.35% | 97.51% | 48.53% | 64.80% | 0.9567 | 0.8074 |

Table7 — Federated Performance with Different Client Counts

Table 7 reports federated performance with 3, 5, and 10 clients under IID and non-IID partitioning.

Under IID partitioning, adding clients barely changed the results. Accuracy stayed between 95.49% and 95.51%, and precision, recall, F1-score, ROC-AUC, and PR-AUC shifted only slightly. This is consistent with the roughly equal class proportions across IID clients: within the tested range, the number of clients had little effect on aggregate performance.

The non-IID results varied more, and the client distributions behind them are uneven. As Fig. 2 shows, diabetic-class proportions range from 0% to 95.63% across the ten clients, and several clients contain no diabetic samples. With three clients, the model reached 95.65% accuracy, 80.31% precision, 67.10% recall, and 73.11% F1-score. At five clients, precision dropped to 74.74% and F1-score to 71.14%, while recall edged up to 67.87%. At ten clients, precision jumped to 97.51% but recall fell to 48.53%, giving an F1-score of 64.80%. Fig. 3 shows this shift most clearly at

ten clients. At the common threshold, that model is more conservative: it makes fewer false-positive predictions but misses more diabetic cases.

The client distribution is not the whole explanation. For the same ten-client non-IID configuration, the 0.50 threshold gave 70.04% precision and 70.99% recall, while the common threshold of 0.8752 gave 97.51% precision and 48.53% recall. The higher threshold therefore accounts for a large part of the precision gain and recall loss, alongside the uneven class distributions across clients.

Those distributions are very uneven. Fig. 3 shows diabetic-class proportions from 0% to 95.63% across the ten clients, and several clients contain no diabetic samples. Changing the number of clients also changes the Dirichlet partition, the client composition, and the client sample sizes, so these experiments cannot separate the effect of client count from those changes. The results describe how these particular generated configurations behaved. They do not show that client count alone causes the differences.

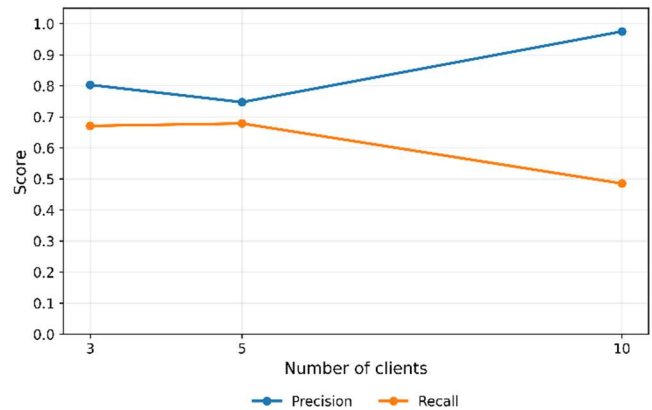
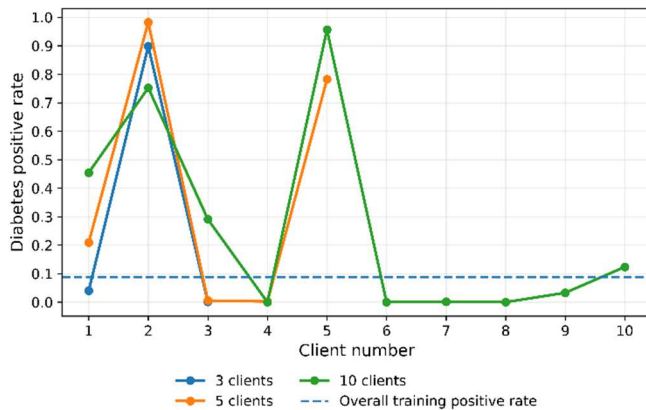


Fig. 2. Non-IID client class distributionsFig. 3. Precision-recall trend across Non-IID client counts

| Non-IID Clients | TP    | FP  | TN     | FN  |
|-----------------|-------|-----|--------|-----|
| 3               | 1,138 | 279 | 17,255 | 558 |
| 5               | 1,151 | 389 | 17,145 | 545 |
| 10              | 823   | 21  | 17,513 | 873 |

Table8 —Non-IID Classification Counts

Table8 gives the classification counts behind the non-IID results. At the common threshold, the ten-client model produces only 21 false positives, against 279 for three clients and 389 for five. That drop accounts for its precision of 97.51%. The same model also produces 873 false negatives, against 558 and 545, so it finds fewer diabetic cases and its recall falls to 48.53%. The high precision of the ten-client model therefore comes with many more missed diabetic cases. It is not an improvement across all measures.

## V. CONCLUSION

This paper compared centralized and federated logistic regression for diabetes prediction, with a focus on how client-level statistical heterogeneity affects federated performance. With three clients, the federated model performed comparably to the centralized baseline under both IID and non-IID partitioning, even though the non-IID clients held diabetic-class proportions from 0.09% to 89.86%. Under IID partitioning, performance also stayed stable as the number of clients grew from 3 to 10. Under non-IID partitioning, precision and recall varied more with client count. At ten clients, precision rose to 97.51% while recall fell to 48.53%, and part of that shift comes from the shared decision threshold in addition to the client distributions. Because changing the client count also changes the generated partition, these differences cannot be attributed to client count alone. Overall, the results support federated

learning as a way to build a diabetes prediction model across institutions without sending clients' raw training data to a central server. The clients here were simulated by partitioning a single dataset, so these findings still need to be checked on real multi-institution data.

Future research in this project would expand into multiple directions as follows: use of differential privacy for better data privacy, application of multi-task federated learning for multi-disease diagnosis and development of federated optimization that can be used in a heterogeneous client setting. Another important aspect to consider in the future would be a dashboard for overall evaluation of the system.